Name of the Student	SHAM S.
Reg. No	192325076
GitHub Link	https://github.com/Sham-2005/MLA0403-Deep_learning

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 2

Game Based Learning

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA04**

SLOT :

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25% **Duration: 1 Hour**

Marks: 50

Assessment Tool Weightage: 35% ***Code of Conduct:***

I SHAM S. (192325076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Name of the student:**

Criteria	Understanding of Concept **(2.5)**	Conceptual Clarity **(2.5)**	Problem-Solving Ability **(2)**	Solution Design & Application **(2)**	Teamwork & Game Participation ***(1)***	**Total (10)**
Weightage	***25 %***	***25%***	***20%***	***20 %***	***10 %***	**100%**
Round 1						
Round 2						
Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS:01 **Total Marks: 50**

Question:

Design and participate in a two-round game-based learning activity to explore the fundamental concepts of Convolutional Neural Networks (CNNs), including architecture, layers, filters, parameter sharing, regularization, AlexNet, ResNet, and real-world applications. Progress through five levels in each round, applying your knowledge to solve challenges involving CNN design, feature extraction, architecture comparison, and practical applications.

The Game Progression

Round 1:

Basic CNN Concepts → Filters → Parameter Sharing → Architecture → AlexNet vs ResNet

Round 2:

Convolution → Feature Maps → Regularization → Architecture Comparison → RealWorld Application

ROUND 1: CNN Architecture Builder (Learn by constructing and understanding CNN architectures layer by layer)

Level	Challenge	Description
Level 1	**CNN Layer Matcher**	Match CNN layers such as Convolution, Pooling, Flatten, and Fully Connected layers with their respective functions and roles in the network.
Level 2	**Filter Finder**	Identify suitable filters and kernels for feature extraction and understand how filter size, stride, and padding affect the output feature map.
Level 3	**Parameter Sharing Puzzle**	Determine how the same filter weights are reused across different image locations and calculate the number of parameters in a convolutional layer.
Level 4	**CNN Architecture Debugger**	Identify and correct problems in a CNN architecture, such as incorrect layer order, incompatible dimensions, and inappropriate layer selection.
Level 5	**AlexNet vs ResNet Challenge**	Compare AlexNet and ResNet architectures and identify how their architectural approaches differ, particularly the use of residual connections in ResNet.

ROUND 2: CNN Feature Detective (Learn by exploring how CNNs extract features, avoid overfitting, and solve real-world problems)

Level	Challenge	Description
Level 1	**Edge Detection Explorer**	Apply simple (3\times3) filters to an image and determine how convolution operations can detect edges and other basic features.
Level 2	**Feature Map Detective**	Analyze feature maps produced by different CNN layers and identify the progression from simple features such as edges to complex patterns and objects.
Level 3	**Regularization**	Identify overfitting in a CNN model and select suitable

	Rescuer	regularization techniques to improve generalization and model performance.
Level 4	**CNN Architecture Race**	Compare ResNet and AlexNet for different image-classification applications and determine which architecture is more suitable for a given problem.
Level 5	**CNN Application Master**	Design a CNN-based solution for a real-world application such as medical image analysis, face recognition, object detection, or autonomous driving, and justify the selected architecture and layers.

Good morning everyone. Today I am going to talk about **Filter Finder**, which is an important concept in Convolutional Neural Networks, or CNNs.

The main purpose of Filter Finder is to identify **suitable filters and kernels for feature extraction** from an input image. A CNN uses small matrices called **kernels or filters** that slide over an image and detect important patterns. For example, a filter can detect edges, corners, textures, or more complex shapes.

The first important factor is **filter size**. Common filter sizes are 3×3, 5×5, and 7×7. A smaller filter such as 3×3 can capture fine details and requires fewer computations. Larger filters can capture a wider area of the image but require more computation.

The second factor is **stride**. Stride determines how many pixels the filter moves at each step. With a stride of 1, the filter moves one pixel at a time and produces a larger feature map. With a larger stride, such as 2, the filter moves faster and produces a smaller feature map.

The third factor is **padding**. Padding adds extra pixels, usually zeros, around the border of the input image. Without padding, the feature map becomes smaller after convolution. Using appropriate padding, such as "same" padding, helps preserve the spatial dimensions of the input.

The output feature-map size can be calculated using the formula:

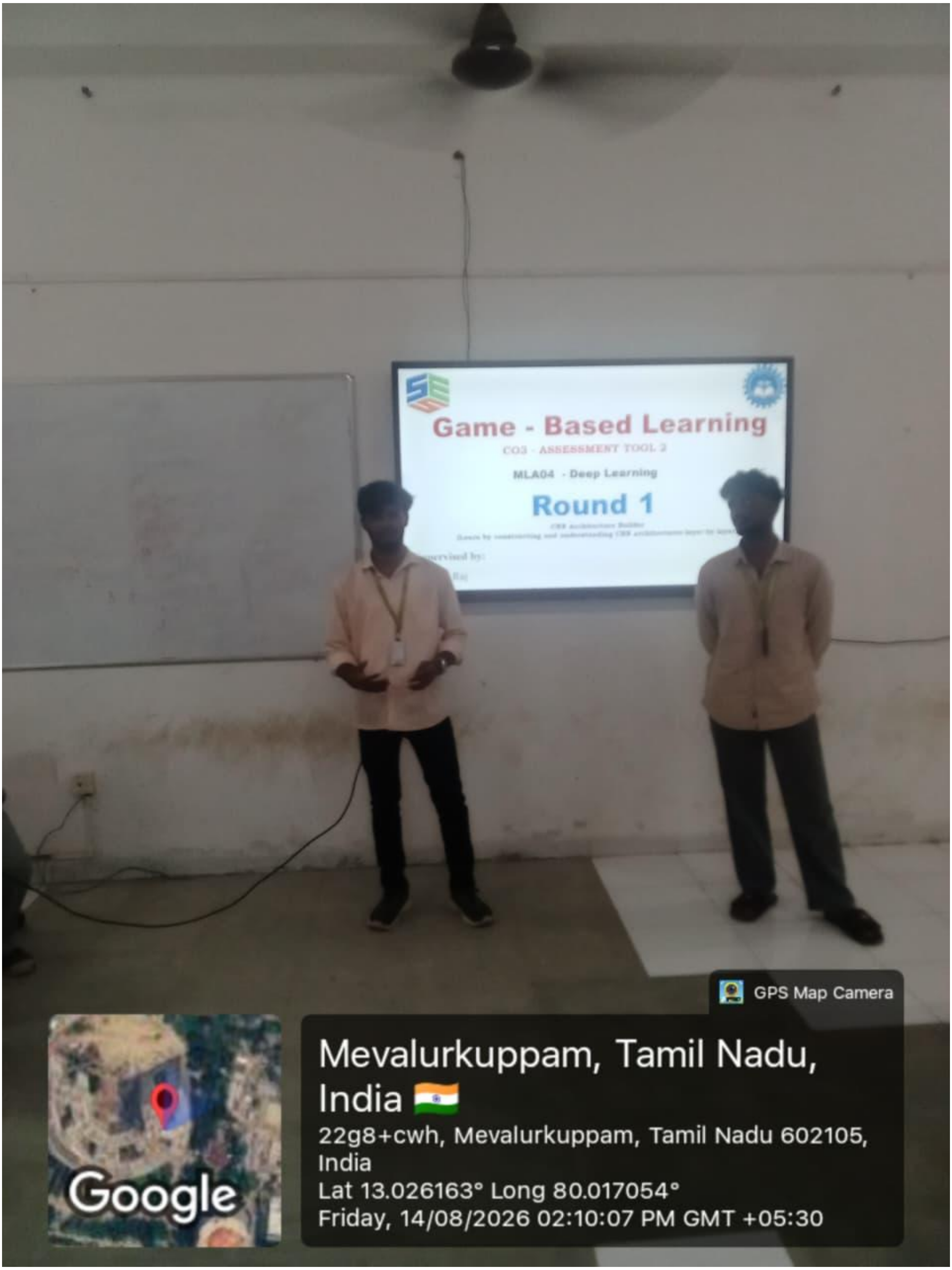
Output size = ((Input size − Filter size + 2 × Padding) / Stride) + 1.

For example, if the input is 32×32, the filter is 3×3, padding is 1, and stride is 1, the output will remain **32×32**.

Therefore, choosing the right filter size, stride, and padding is important because they directly affect the **features extracted, computational cost, and size of the feature map**.

In conclusion, Filter Finder helps us understand how CNN filters extract meaningful information from images and how convolution parameters control the final feature representation.

Thank you.


Game - Based Learning
CO3 - ASSESSMENT TOOL 2
MLA04 - Deep Learning
Round 1
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026163° Long 80.017054°
Friday, 14/08/2026 02:10:07 PM GMT +05:30
Google

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Conceptual Clarity	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Problem-Solving Ability	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design & Application	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Teamwork & Game Participation	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand